Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – IX

Roll No.: S021	Name:LODHI IQRA
Class:SYBsc IT	Subject: DATA STRUCTURE
Date of Assignment:19/08/2026	Date/Time of Submission:19/08/2026

PART B - QUESTIONS

Searching Algorithms: Write programs to implement and compare:

a. Linear search.

<u>CODE:</u>

```
# Linear Search in Python


def linearSearch(array, n, x):

    # Going through array sequencially
    for i in range(0, n):
        if (array[i] == x):
            return i
    return -1


array = [2, 4, 0, 1, 9]
x = 1
n = len(array)
result = linearSearch(array, n, x)
if(result == -1):
    print("Element not found")
else:
    print("Element found at index: ", result)
print("IQRA S021")
```

<u>OUTPUT:</u>

```
================= RESTART: D:/D
Element found at index:  3
IQRA S021
```

b. Binary search (on a sorted array).

CODE:

```python
# Binary Search in python


def binarySearch(array, x, low, high):

    # Repeat until the pointers low and high meet each other
    while low <= high:

        mid = low + (high - low)//2

        if x == array[mid]:
            return mid

        elif x > array[mid]:
            low = mid + 1

        else:
            high = mid - 1

    return -1


array = [3, 4, 5, 6, 7, 8, 9]
x = 4

result = binarySearch(array, x, 0, len(array)-1)

if result != -1:
    print("Element is present at index " + str(result))
else:
    print("Not found")

print("IQRA S021")
```

OUTPUT:

```
================= RESTART: D:/DAT
Element is present at index 1
IQRA S021
```